

EMPLOYMENT & WAGE DYNAMICS · ECI ANALYTICS

NAICS 3344 Semiconductor Sector

Dashboard Technical Documentation

ECI Employment Analytics · Semiconductor & Electronic Component Manufacturing

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U.S. Bureau of Labor Statistics · Employment Cost Index

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1. Overview

This document describes the NAICS 3344 Semiconductor Sector Employment & Wage Dynamics Dashboard, an interactive analytical report built in Power BI for the Employment Cost Index (ECI) Analytics series. The dashboard tracks 365,200 workers across the semiconductor and electronic component manufacturing sector as of December 2025, surfacing wage premium trends, skill-based pay gaps, employment momentum, and real versus nominal growth across all NAICS 334 subsectors.

The dashboard investigates what the data calls the Semiconductor Paradox: despite semiconductors powering AI, defense, and modern infrastructure, U.S. sector employment declined in 2025. Yet wages tell the opposite story — a +21.9% premium over the manufacturing average, still widening. This report documents the full analytical workflow, from raw BLS data extraction through Python, to CSV/Excel intermediary files, through to the final polished Power BI dashboard.

Key Metrics Snapshot — December 2025

Employment 365.2K ▼ 4.6% YoY	Wage Premium +21.9% +1.6 pp expansion	Skills Premium 0.3% ▼ 0.1 pp compressed	Wage Acceleration +6.9% Q4 sprint +0.56%
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2. Data Sources

All metrics are sourced from the U.S. Bureau of Labor Statistics Employment Cost Index series for NAICS 3344. Data is published quarterly and reflects compensation cost changes for civilian workers.

2.1 BLS Source Tables

Variable	File	Adjustment	Scope
Employees_sa	Table B-1a Employment and Earnings.pdf	SA	All employees
weekly_sa	Table B-3a average_weekly_earnings.pdf	SA	All employees
weekly_nsa	Table B-3a average_weekly_earnings_not_seasonally_adjusted.pdf	NSA	All employees
hourly_sa	Table B-3b average_hourly_earnings.pdf	SA	All employees
hourly_nsa	Table B-3b average_hourly_earnings_not_seasonally_adjusted.pdf	NSA	All employees
weekly_prod_sa	Table B-8a average_weekly_earnings_production_employees.pdf	SA	Production workers
weekly_prod_nsa	Table B-8a ...not_seasonally_adjusted.pdf	NSA	Production workers
hourly_prod_sa	Table B-8b average_hourly_earnings_production_employees.pdf	SA	Production workers
hourly_prod_nsa	Table B-8b ...not_seasonally_adjusted.pdf	NSA	Production workers

2.2 Analytical Workflow

All files were originally downloaded as PDFs from BLS. They were then meticulously converted into CSV format, with a focus on NAICS 334. The data pipeline followed a three-stage process before the dashboard was built:

- Stage 1 — Data Extraction: Eight BLS PDF files were loaded into Python using pandas.
 - Six custom insight functions processed the raw tables, calculating wage growth rankings, sector premiums, stagnation risk, skills gaps, temporal acceleration, and real purchasing power adjustments.
- Stage 2 — Intermediary Files: Python exported six CSV output files and two Excel workbooks to a dedicated Findings subfolder. These files serve as the clean, validated data layer that feeds the Power BI data model, ensuring full reproducibility and a clear audit trail from source to visual.
- Stage 3 — Dashboard Design & Build: With the data layer confirmed, the dashboard layout, color palette, DAX measures, and HTML/CSS/JS custom visuals were planned and built in Power BI Desktop.

2.3 Python Output Files

File	Source Function	Format
insight_1_wage_growth_rankings.csv	insight_1_communications_surge()	DataFrame (direct)
insight_2_semiconductor_premium.csv	insight_2_premium_gap()	Single-row DataFrame
insight_3_stagnation_risk.csv / .xlsx	insight_3_stagnation_risk()	DataFrame (direct)
insight_4_skills_premium.csv	insight_4_skills_premium()	Single-row DataFrame
insight_5_recent_acceleration.csv	insight_5_recent_acceleration()	Single-row DataFrame
insight_6_real_vs_nominal.csv	Rebuilt loop in output block	Multi-row DataFrame
employment_trajectory_by_month.xlsx	Table B-1a employment_data.csv	Excel workbook
Weekly_wage_trajectory_by_month.xlsx	Table B-3a / B-3b	Excel workbook

3. Dashboard Design

The dashboard was designed to a polished, production-grade standard, going well beyond native Power BI components. A significant portion of the UI/UX experience is delivered through custom HTML, CSS, and JavaScript visuals embedded directly inside Power BI, complemented by DAX measures for all calculated metrics and KPIs.

3.1 Design Philosophy

The design goal was to create a refined analytical experience that communicates a complex labour market story clearly and compellingly. Every visual decision — from typography to color usage to spacing — was made to serve the data, not decorate it. The result is a dashboard that reads as both a rigorous analytical tool and a polished executive-facing report.

3.2 Color Palette

Color	Hex	Usage
Dark Forest Green	#0A3D20	Primary background, sidebar, section headers
Emerald Green	#00C96B	Positive values, KPI highlights, accent lines
Light Mint	#E8F5EE	Card backgrounds, alternating table rows
Deep Red	#D93025	Negative values, wage loss indicators, alerts
Amber / Orange	#E67E22	Caution signals, skills compression, risk flags
Off-White	#F4F9F6	Body text areas, clean panel backgrounds
Muted Green-Gray	#5A7A62	Secondary labels, metadata, subtitles

3.3 Technical Stack

The dashboard was built using a hybrid technical stack that combines native Power BI components with custom-coded visuals to deliver a richer, more polished user experience:

- **DAX (Data Analysis Expressions):** All calculated metrics, KPIs, year-over-year comparisons, premium calculations, real wage adjustments, and time-intelligence measures are implemented as DAX measures and calculated columns within the Power BI data model. This ensures every figure is live, dynamically recalculated, and tied directly to the source data.
- **HTML + CSS (Custom Visuals):** KPI cards, insight panels, the Executive Summary, the Storytelling panel, the Executive Walkthrough, and the About This Dashboard section, the diverging bar chart for wage growth by segment, and the real vs. nominal comparison table are built as HTML Content visuals inside Power BI. Custom CSS delivers the typography, card layouts, color-coded indicators, animated badge chips, and responsive spacing that native visuals cannot provide.
- **JavaScript:** Interactive behaviors — including the guided walkthrough step logic, tooltip popover content, panel open/close animation, and dynamic content rendering — are implemented in JavaScript within the HTML visual layer.
- **Native Power BI Visuals:** The employment trend line chart is built using native Power BI visuals styled to match the custom color palette.

4. Metrics — Definition & Rationale

Every metric on the dashboard was chosen to answer a specific analytical question about the semiconductor sector workforce. The table below documents each metric, its calculation, and the reason it was included.

Metric	Value (Dec-25)	Calculation	Why It Matters
Sector Employment	365.2K workers	BLS Table B-1a seasonally adjusted headcount for NAICS 3344	Baseline workforce size; tracks whether the sector is growing or contracting in headcount terms
YoY Employment Change	-4.6% / -17.7K jobs	$(\text{Dec-25} - \text{Dec-24}) / \text{Dec-24} \times 100$	Measures the structural direction of hiring; a persistent negative signals automation or offshoring pressure
Monthly Employment Change	-100 jobs (Nov→Dec)	Dec-25 employees - Nov-25 employees	Short-term momentum indicator; near-zero monthly change confirms Q4 stabilization plateau
Wage Premium vs Mfg Avg	+21.9% / +\$21.8K/yr	$(\text{Semiconductor weekly} - \text{Mfg avg weekly}) / \text{Mfg avg} \times 100, \text{ annualised } \times 52$	Quantifies the pay advantage that semiconductor work commands; a widening premium signals the sector is competing hard for talent
Premium Expansion YoY	+1.6 pp	Premium% Dec-25 - Premium% Dec-24	Directional signal; if the gap widens while headcount falls, employers are paying more per worker to retain a smaller, more specialised workforce
Skills Premium (White/Blue Collar Gap)	0.3% (from 0.4% in 2024)	$(\text{All-employee weekly} - \text{Production worker weekly}) / \text{Production worker weekly} \times 100 \text{ for NAICS 3344}$	Reveals whether the pay gap between knowledge workers and production workers is expanding or narrowing; compression can signal oversupply of specialised talent
Nominal Wage Growth (Full Year)	+6.9%	$(\text{Dec-25 weekly} - \text{Dec-24 weekly}) / \text{Dec-24 weekly} \times 100 \text{ for NAICS 334413}$	Headline compensation growth figure; benchmarked against the broader manufacturing average to assess sector competitiveness
Q4 Wage Acceleration	+0.56% in Q4 alone	$(\text{Dec-25} - \text{Oct-25}) / \text{Oct-25} \times 100 \text{ for NAICS 334413}$	Decomposes annual growth into early-year vs. late-year momentum; a Q4 sprint suggests employers accelerated raises to retain workers heading into year-end
Real Wage Growth	Varies by subsector	Nominal growth % - 3.5% inflation assumption (Fisher approximation)	Adjusts headline growth for purchasing power; distinguishes genuine income gains from inflation-eroded raises. Workers with

			real growth below 0% are losing ground in real terms
Subsector Real Wage Ranking	Comms Gear +8.52% best; Electronic Parts -3.54% worst	Real growth calculated per subsector using Dec-24 and Dec-25 weekly earnings	Surfaces divergence within the sector; the 12-percentage-point spread between best and worst subsectors reveals that NAICS 334 is not one homogeneous labour market

5. Key Findings

The six Python insight functions collectively profile U.S. semiconductor-sector earnings dynamics across the December 2024 to December 2025 period. The dashboard surfaces these findings as its core narrative.

Finding 1 — Employment Contraction with Stabilization

The sector shed 17,700 jobs in 2025 (−4.6% YoY), stabilizing at 365,200 workers by December. Critically, the decline is structural rather than shock-driven: the front-loaded losses of Dec'24–Aug'25 (approximately −12,500 jobs) gave way to a near-plateau in Q4, with only −100 jobs in Nov→Dec. Three consecutive months within a 200-job range confirm the contraction phase has ended. This signals that automation and offshoring — not a demand shock — drove the reduction.

Finding 2 — Expanding Wage Premium

Semiconductor employees earn 21.9% above the manufacturing average — a premium of \$21,800 per year. This gap widened by +1.6 percentage points in 2025. The simultaneous contraction in headcount and expansion in wage premium is the central paradox the dashboard investigates: the sector is paying more per worker to retain fewer, more specialised employees. This is consistent with a shift from growth-by-headcount to growth-by-value-per-worker.

Finding 3 — Accelerating Wage Growth

Nominal wages rose +6.9% in 2025 for semiconductor device workers (NAICS 334413), with a Q4 sprint of +0.56% in the final two months alone. The Q4 acceleration suggests employers intensified retention efforts heading into year-end, reinforcing the hypothesis that retention — not recruitment — is the dominant labour dynamic in the current cycle.

Finding 4 — Skill Compression Risk

The wage edge for highly skilled (non-production) roles narrowed from 0.4% to 0.3% in 2025. This compression suggests either an oversupply of specialised talent or weakening employer demand for top-tier skills — possibly reflecting automation absorbing some high-skill tasks. If this trend continues, it may reduce the sector's attractiveness as a premium destination for knowledge workers.

Finding 5 — Subsector Divergence

The headline sector figures mask dramatic divergence at the subsector level. Communications Gear (NAICS 3342) posted +8.52% real wage growth in 2025, driven by surging demand for networking and telecom infrastructure. Electronic Parts (NAICS 334416) posted −3.54% real growth — meaning workers there lost purchasing power. Two of five subsectors posted real wage losses. This divergence exposes vulnerabilities that aggregate sector statistics conceal, and was a key rationale for including subsector-level breakdowns in the dashboard.

Finding 6 — Hypotheses Tested

The dashboard explicitly surfaces four analytical hypotheses and their verdicts:

- REJECTED — Demand drives headcount: Global chip demand did not produce U.S. job growth — automation and offshoring absorbed the expansion.
- CONFIRMED — Premiums signal scarcity: Widening skills gap (+0.3%) and Q4 wage sprint (+0.56%) confirm competition for specialized talent.
- CONFIRMED — Stabilization underway: Three months near 365K with near-zero delta confirm the contraction phase has ended.
- REJECTED — Sub-sectors move together: Comms Gear (+8.52% real) vs. Electronic Parts (-3.54%) reveal stark divergence within NAICS 334.

6. Control Panel & Navigation

The left-side navigation panel provides five distinct interaction modes, each opening a dedicated full-screen overlay panel. These panels transform the dashboard from a static report into a guided analytical experience. All panels are built as HTML Content visuals inside Power BI, styled with custom CSS and animated with JavaScript.

6.1 Executive Summary

A concise, practical overview designed for fast consumption by decision-makers who need the bottom line quickly. The panel displays the three headline metrics (Employment, Wage Premium, Wage Growth), three numbered Key Findings with tagged chips, a Real Wage Growth by Subsector mini-chart, two Emerging Risk cards (Skill Compression and Subsector Divergence), and a Strategic Recommendation with action tags (RETENTION · UPSKILLING · PAY DIFFERENTIATION · TALENT MOBILITY). The panel closes via an X button in the top-right corner.

6.2 Storytelling

A deeper narrative walkthrough of the data that goes beyond summary bullets to explain the why behind the trends. This panel is titled "The Semiconductor Paradox — Employment & Wage Dynamics" and presents the full analytical narrative in three acts: Act I (Contraction), Act II (Premium Widens), and Act III (Retention Signal). It includes a Hypotheses Tested section with four cards marked CONFIRMED or REJECTED, a direct link to the official BLS website for source verification, and a footer citing the data sources and ECI quarter. This panel is designed for analysts and stakeholders who want to understand the story the data is telling, not just the numbers.

6.3 Executive Walkthrough

A chart-by-chart guided tour of the entire dashboard, structured as a 9-step interactive walkthrough. Each step highlights one visual element — pointing to it with an arrow indicator — and explains what it measures, how to read it, and what the current data signals. Steps are navigated using Next and Close buttons. The walkthrough is designed for analysts and stakeholders who want a thorough understanding of every panel without requiring prior familiarity with ECI methodology or Power BI navigation. The 9 steps cover: (1) Welcome, (2) Employment KPI, (3) Wage Premium KPI, (4) Skills Premium KPI, (5) Wage Acceleration KPI, (6) Wage Trends bar chart, (7) Employment Trend line, (8) Real vs Nominal table, (9) Closing summary.

6.4 Data Source

A navigation button that opens a direct URL to the official BLS website where the underlying datasets are publicly available. This link points to the Employment Cost Index tables used to build the dashboard — providing full transparency and enabling independent verification of all figures presented. The BLS tables referenced are: B-1a SA, B-1b NSA, B-3a SA, B-3b NSA, B-8a SA, and B-8b NSA.

6.5 About This Dashboard

A documentation panel that covers the purpose, data sources, methodology, and technical stack behind the dashboard. It also documents known UI behaviors (specifically, the semi-transparent white overlay that can appear after opening and closing HTML panels — a known behavior of HTML Content visuals in Power BI, resolved by reloading the page), design decisions, and how to correctly interpret the metrics presented throughout the report. The panel is divided into sections: Known Behavior, Design Philosophy, Dashboard Navigation Guide (describing all 6 panel types), and Overview/Purpose/Data & Interactivity/Technical Stack.

6.6 LinkedIn & Copyright

A button in the bottom-left corner of the control panel navigates directly to the author's LinkedIn profile at [linkedin.com/in/aaronobando](https://www.linkedin.com/in/aaronobando), where the full copyright notice is located. The dashboard footer also displays: "Built by Aaron E. Obando · ECI Analytics" alongside a link to the official BLS ECI Q4 2025 release. All analytical content, dashboard design, DAX measures, and custom HTML/CSS/JS visuals are the original work of Aaron E. Obando.

7. UI/UX Design

The dashboard was built to a production-grade UI/UX standard that goes well beyond what is typically seen in Power BI reports. The design intent was to create an experience that feels as polished as a fintech or analytics SaaS product, not a default Power BI template.

7.1 Visual Hierarchy

The layout follows a clear three-tier hierarchy: (1) the left navigation panel anchors the experience and provides persistent access to all interactive panels; (2) the top KPI row delivers the four headline metrics at a glance, each in a card with a colored badge indicator; (3) the main body area contains the analytical visuals — the wage trend bar chart, the employment trend line, and the real vs. nominal comparison table. This hierarchy guides the reader from summary to detail without requiring instruction.

7.2 KPI Cards

Each of the four KPI cards follows a consistent structure: a label (NAICS code and metric name), a large headline value, a colored trend badge (green for positive, red for negative), a plain-English description line, and a secondary metadata line. Badge colors are semantically meaningful and consistent across the entire dashboard — green always means positive/gain, red always means negative/loss, amber signals caution or compression.

7.3 Interactive Tooltips

Both the wage trend bar chart and the real vs. nominal table support hover tooltips. Hovering over any bar reveals a popover card with the subsector name, a headline growth figure, and a one- to two-sentence contextual explanation of the drivers behind the figure. This contextual layer is implemented in the HTML visual layer using JavaScript event listeners and CSS-positioned overlay divs.

7.4 Known UI Behavior

After opening a panel such as the Executive Summary, Executive Walkthrough, or the About panel, clicking a hyperlink inside may leave a semi-transparent white overlay that blocks interaction with the rest of the dashboard. This is a known behavior of HTML Content visuals in Power BI, caused by a white semi-transparent shape layer backing the HTML visual. When a hyperlink is followed, that overlay layer can remain on screen and intercept clicks. The resolution is to reload the page to fully restore dashboard interactivity. This behavior is documented in the About This Dashboard panel.

8. Limitations & Notes

Filtering and drill-down capabilities are constrained by the structure of the source data. The BLS Employment Cost Index publishes NAICS 334 figures at a fixed level of aggregation — granular breakdowns by state, occupation, or firm size are not available in the public release.

- The 3.5% inflation assumption used for real wage calculations is a modelling constant. When official CPI figures are finalised for 2025, this assumption should be updated and the real wage figures recalculated accordingly.

- The `insight_6_real_vs_nominal()` Python function contains known technical debt: the output loop is duplicated in the script's output block because the function only prints rather than returning a DataFrame. This is flagged as a refactoring candidate.
- Data spans August through December 2025 for employment trajectory, and December 2024 through December 2025 for wage comparisons. Figures are not directly comparable to periods before December 2024 without additional historical files.

9. Summary

The NAICS 3344 Semiconductor ECI Dashboard delivers a complete analytical picture of U.S. semiconductor workforce dynamics for 2025. Starting from eight raw BLS CSV files, a six-function Python pipeline extracted and validated all analytical outputs, which were then exported to clean CSV and Excel intermediary files. The Power BI dashboard was built on top of these files using a hybrid stack of DAX measures, native visuals, and custom HTML/CSS/JavaScript panels.

The result is a production-grade, UI/UX-polished dashboard that tells the story of the Semiconductor Paradox — a sector simultaneously losing headcount and gaining wage power — through four KPI cards, two analytical charts, a multi-sector comparison table, and five interactive navigation panels. Every metric is defined, every finding is substantiated, and every visual decision is deliberate.

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U.S. Bureau of Labor Statistics · B-1a SA · B-1b NSA · B-3a SA · B-3b NSA · B-8a SA · B-8b NSA